The Carry-Over Lottery Allocation Family: Bounds, Backtests, and a Configuration Space for NBA Draft Reform

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Abstract

The annual NBA draft assigns 30 indivisible picks to 30 teams. The top picks go through a lottery among the 14 non-playoff teams, which carries a documented manipulation incentive ("tanking"): a team eliminated from playoff contention can improve its expected draft position by losing more games. Banchio and Munro [1] prove that no mechanism based on end-of-season records alone can simultaneously reward poor performance and eliminate this incentive. Carry-Over Lottery Allocation (COLA) sidesteps the impossibility by replacing single-season records with multi-year playoff history as the basis for draft priority [2]. We formalize the COLA family as a configuration space parameterized by seven design dials (eligibility, increment, diminishment, lottery weighting, cap, carry-over scope, tiebreak). Each variant proposed in Highley et al. [2] and the Substack series is a point in this space, and the NBA's 2026 "3-2-1" proposal sits at a degenerate corner. We instantiate the framework in the full ZenGM Basketball-GM engine, run headless with the per-game simulation active (generated rosters and contracts, AI general managers, city-size effects), across 5,220 league-seasons, so team strength emerges from the rosters the engine plays rather than from a stipulated law of motion. The discriminating axis is anti-tanking, measured by the single-season tanking gradient (the worst-versus-fifth-worst expected-pick gap ranked by current record; about zero is the target). Classic leaves a $+2.48 \pm 0.11$ gradient (twenty-two standard deviations from zero: losing pays) because it lotteries only the top four picks and orders the rest by record. Full-depth proportional weighting ($\gamma = 1$), which draws every pick by the multi-year index, closes that tail to $+0.29 \pm 0.33$ (indistinguishable from zero) while directing help to the long-droughted more sharply than any alternative (-0.94 ± 0.02 picks per drought-year). This is a genuine tradeoff: full-depth weighting removes the residual tanking but gives up the top-four lottery Classic keeps to line up with the current NBA system, a deliberate familiarity choice in Highley et al. [2]. We present $\gamma = 1$ as the weighting that clears the anti-tanking test, flag that tradeoff, and characterize a middle option that keeps the four-pick lottery while ordering picks 5–14 by the index. We also derive a closed-form $\sim 4\%$ upper bound on the worst-case probability gain from pool manipulation in the Classic configuration (Theorem 2) and a per-series tanking bound for the Capped configuration (Lemma 3). Simulating the pool-swap loophole on the engine's pre-draft index confirms the bound: the worst-case realized gain under $\gamma = 1$ is at most 2.67% in the 14-team pool and 1.54% in the 22-team pool, and the anti-correlation premise behind it holds in the engine. A 26-season historical backtest (1999–2025) reproduces the Philadelphia "Process" as a second-tier validation. All numbers rest on $n = 48$ independent leagues per mechanism: Classic's gradient is firmly positive, $\gamma = 1$'s 95% confidence interval includes zero (no measurable tanking reward), and the Classic-minus-$\gamma = 1$ contrast is Welch $t = 6.35$, $p < 10^{-6}$, Cohen's $d = 1.30$.

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Simulation parameters. The empirical results are produced by the ZenGM Basketball-GM engine run headless, with the full per-game simulation active: generated rosters and contracts, AI general managers, and city-size effects (draft type COLA). Each mechanism is evaluated over 48 independent leagues (seeded per replicate), each simulated for 15 consecutive seasons. Uncertainty is reported at the replicate level, so season-to-season correlation within a league does not inflate significance; the effective sample size for any mechanism comparison is 48, not the raw season count. The full study spans 5,220 league-seasons (156,600 team-seasons): a draft-to-outcome channel probe (12 leagues, $n = 12$) and a lottery-weighting sweep over seven mechanisms (7×48 leagues), both at 15 seasons each. The pool-manipulation analysis re-uses the sweep's instrumented runs and adds no further seasons.

1 Introduction

In the 2025-26 NBA season, by some counts roughly a third of the league's 30 teams were openly losing games at season's end to improve their lottery odds. The league responded with several reform ideas, including the "3-2-1 lottery" [3]. The proposal carries a three-year re-evaluation provision: the design discussion will reopen in 2029.

The annual NBA draft assigns thirty indivisible draft picks to thirty teams. Picks 15–30 go by reverse regular-season record. Picks 1–14 go through a lottery confined to the 14 non-playoff teams, with weighted odds tilted toward teams with worse records (in the current system, the bottom three teams are weighted equally and the rest decline). The NBA has adjusted those odds repeatedly while facing the same question: how to assign indivisible picks while (a) preferentially rewarding the weakest teams, (b) eliminating the incentive to under-perform, and (c) keeping the mechanism transparent enough that fans, owners, and players can reason about it.

Banchio and Munro [1] give a formal treatment of the central tension and prove an impossibility. They define a *No-Tanking Condition* (NTC): no team can improve its expected outcome by losing. Any mechanism mapping end-of-season records to draft probabilities while giving worse-record teams strictly better odds must violate NTC. Only a uniform lottery satisfies NTC, abandoning differential reward. The 2026 "3-2-1" proposal takes precisely this route, with adjustments for the play-in tournament and reduced odds for the three teams at the bottom [3].

Carry-Over Lottery Allocation (COLA) sidesteps the impossibility on a different axis: it replaces single-season records with multi-year playoff history as the basis for draft priority [2]. Each non-playoff team accumulates draft priority over consecutive seasons; success in the playoffs or the draft itself diminishes that priority. The single-season manipulation channel that drives the Munro-Banchio counterexample becomes ineffective: no single-season choice materially shifts the multi-year ticket pool. COLA is a family rather than a single design; the original paper develops Classic COLA, and subsequent work introduces variants at different points in the design space.

Contributions:

- **(§3) Seven-dial configuration space.** We formalize the COLA family as a 7-tuple $(E, \Delta, \rho, W, C, S, T)$: eligibility, increment, diminishment, lottery weighting, cap, carry-over scope, tiebreak (Definition 2). Each named variant (Simple, Simple Lottery, Classic, Countdown, Capped) is a point in this space; the NBA's 2026 "3-2-1" proposal sits at a degenerate corner. Each dial maps to a downstream league-relevant property, so variant comparison becomes tuple comparison.
- **(§5) Full-engine evaluation and the $\gamma = 1$ weighting.** We evaluate the lottery-weighting dial in the full ZenGM Basketball-GM engine (per-game simulation, generated rosters and contracts, AI general managers, city-size effects), run headless across 5,220 league-seasons. Sweeping the weighting as a

one-knob family (draw the full draft order with probability proportional to the multi-year index raised to a power γ) isolates the dial responsible for anti-tanking. Full-depth proportional weighting ($\gamma = 1$) flattens the single-season tanking gradient to about zero (from Classic's $+2.48$) while directing draft help to the long-droughted most sharply of any mechanism. It is a one-dial change from Classic on the weighting, but it gives up the top-four lottery Classic keeps for NBA familiarity, so we present it as a tradeoff rather than a free win, and we characterize a middle option (top-four lottery, picks 5–14 by index) that may keep both.

- **(§4, §5) Analytical bounds and empirical confirmation.** A closed-form $\sim 4\%$ upper bound on pool-manipulation gain under reasonable index distributions (Theorem 2), plus a per-series tanking bound for Capped (Lemma 3). Corollary 4: the cap value C drops out of the Capped manipulation bound; only $|E|$ matters. We confirm the bound empirically by simulating the pool-swap loophole on the engine's pre-draft index: the worst-case realized gain under $\gamma = 1$ is at most 2.67% (14-team pool) and 1.54% (22-team pool), both well under the $\sim 4\%$ ceiling, and the bound's structural-anti-correlation premise holds in the engine. The 1999–2025 historical backtest with the 2013–2017 Philadelphia case study is a second-tier validation.

2 Background and Related Work

2.1 Effort Manipulation: The Munro-Banchio Impossibility

Banchio and Munro [1] prove that any mechanism mapping end-of-season records to draft probabilities, while giving the worst team strictly better odds, must violate NTC. The proof constructs a minimal counterexample: two eliminated teams with identical records and a final game between them. The loser finishes with a worse record and therefore receives better draft odds; both teams prefer to lose. Only the uniform lottery satisfies NTC, but uniform odds abandon differential reward of the weakest teams.

The authors propose *T-IC* (Truncated Incentive Compatibility) as an alternative: track each team's conditional probability of finishing last and freeze allocations at the moment IC would first be violated. The mechanism is optimal in theory but has three practical issues. It is opaque to fans; the freeze point depends on team-level playoff-versus-draft valuations the league cannot observe; and the timing assumes teams begin the season with an incentive to win, when some teams tank from the first game.

2.2 Preference Manipulation and Axiomatic Limits

A second impossibility addresses preference misreporting (a team claiming to want a player it does not, to manipulate allocation order): Coreno and Balbuzanov [4] prove that no draft rule simultaneously satisfies Respect for Priority [5], Envy-Free up to One Object [6, 7], and Strategy-Proofness. The result is single-period. Whether the axioms lift to the multi-year priority structure of COLA is open and outside this paper's scope; we note the parallel impossibility as a complement to the Munro-Banchio effort-manipulation channel that motivates the bound in Section 4.

2.3 Prior Reform Proposals

Three reform proposals form the immediate context. The *uniform lottery* (equal odds for all non-playoff teams) is the only mechanism satisfying NTC; the league has not adopted it. The *wheel* (a pre-committed rotation of draft positions, attributed informally to Mike Zarren) eliminates tanking by decoupling record from odds, at the cost of removing the rebuilding mechanism for weak teams. The 2026 *3-2-1 lottery* [8] expands the lottery to 16 teams across four tiers (2:3:2:1 balls per tier from the worst-record tier down) and includes the play-in losers. Structurally it is a uniform lottery with play-in adjustments and a reduction for the bottom three teams, accepting the Munro-Banchio tradeoff. COLA takes a different route: instead of flattening single-season odds, it shifts the priority basis to multi-year playoff history, an axis neither Munro-Banchio nor 3-2-1 considers.

3 The COLA Mechanism Family: A Configuration Space

COLA is a family of mechanisms parameterized by seven design dials. Each NBA lottery reform proposal under recent discussion (the status quo weighted lottery, the 2026 "3-2-1" proposal, and each COLA variant in Highley et al. [2] and the Substack series) corresponds to a specific choice of values for those seven dials. A policymaker who wants to move a downstream property (anti-tanking strength, weak-team rebuild horizon, broadcast complexity, multi-year predictability) is moving one or more dials. We develop the lottery index as the family's unified state variable (§3.1), the seven dials (§3.2), the variant map (§3.3), and the dial-to-property map (§3.4). Incentive compatibility rests on five assumptions [2], summarized in §3.5.

3.1 Lottery Index

Every COLA variant maintains a per-team integer state across seasons and uses it to determine draft priority (DP) in the current season. We call this state the *lottery index* and write $L_i^{(t)}$ for team i at the end of season t. Depending on the variant, $L_i^{(t)}$ tracks drought length (consecutive seasons without a playoff series win or top-3 pick) or a stockpile of lottery tickets; both are manifestations of the same index.

Definition 1 (Lottery Index)**.** *For each team i and season t, the lottery index $L_i^{(t)} \in \mathbb{Z}_{\geq 0}$ is determined by an initial value $L_i^{(0)}$, a per-season* increment $\Delta_i^{(t)}$ *(variant-specific, depending on team performance and eligibility), and a per-season* diminishment $\rho_i^{(t)} \in [0, 1]$ *that scales the index down based on playoff and draft outcomes:*

$$L_i^{(t+1)} = \left(L_i^{(t)} + \Delta_i^{(t)}\right) \cdot \left(1 - \rho_i^{(t)}\right). \tag{1}$$

The pool *in season t is $P^{(t)} = \sum_{i \in E^{(t)}} L_i^{(t)}$, where $E^{(t)} \subseteq \{1, \dots, 30\}$ is the set of teams eligible for the lottery in that season.*

Three dials live in this update: the eligibility set $E^{(t)}$, the increment $\Delta_i^{(t)}$, and the diminishment $\rho_i^{(t)}$. A non-playoff team has positive increment (it accumulates priority); a team that wins a playoff series or a top pick has positive diminishment (it consumes some or all of that priority). §3.2 formalizes these three together with four additional dials that complete the parameterization, and §3.3 populates the 7-tuple for each variant.

3.2 Seven Dials

Four additional dials, implicit in the prose definitions of Highley et al. [2] and the Substack series but not yet formalized as design degrees of freedom, complete the configuration: how draft positions are assigned given the index vector, whether the index is bounded above, how long it persists across seasons, and how ties are resolved.

Definition 2 (COLA Configuration)**.** *A* COLA configuration *is a 7-tuple $(E, \Delta, \rho, W, C, S, T)$ specifying:*

1. ***Eligibility*** *$E^{(t)} \subseteq \{1, \dots, 30\}$: a season-$t$ predicate selecting the teams that participate in the lottery (per Definition 1).*

2. ***Increment*** $\Delta_i^{(t)} \in \mathbb{Z}_{\geq 0}$*: a non-negative integer added to the index of each eligible team, determined by season-t state (per Definition 1).*

3. ***Diminishment*** $\rho_i^{(t)} \in [0, 1]$*: the fraction of the index removed after team i's playoff or draft outcome (per Definition 1).*

4. ***Lottery weighting*** W*: a rule mapping the index vector $L^{(t)}|_{E^{(t)}}$ to a distribution over assignments of teams to draft positions $\{1, 2, \ldots, |E^{(t)}|\}$. The rule may be deterministic (e.g., assign by index rank) or random (e.g., draw pick 1 with probability L_i/P, then draw pick 2 from the remainder).*

5. ***Cap*** $C \in \mathbb{Z}_{>0} \cup \{\infty\}$*: an upper bound on $L_i^{(t)}$, enforced as $L_i^{(t)} \leftarrow \min(L_i^{(t)}, C)$ after each increment. $C = \infty$ denotes no explicit cap.*

6. ***Carry-over scope*** S*: the temporal persistence rule, one of* single-season *(recompute $L_i^{(t)}$ from season-t state alone, no cross-season memory),* unbounded multi-year *(accumulate without limit),* bounded multi-year *(accumulate up to C), or* reset-on-event *(clear the index when a specified event occurs).*

7. ***Tiebreak*** T*: a total ordering on $E^{(t)}$ that resolves cases where $L_i^{(t)} = L_j^{(t)}$ for two or more eligible teams.*

A COLA variant *is a specification of all seven dials.*

Dials 4–7 are not novel mechanisms: each variant in Highley et al. [2] and the Substack series already specifies a choice for each. Promoting them to typed dials enables cross-variant comparison via a single configuration object and dial-by-dial analysis of downstream properties. §3.3 populates the configuration per variant; §3.4 maps dials to league-relevant properties.

3.3 Variant Map

Table 1 populates the seven dials for the six variants under consideration: five COLA variants and the NBA's 2026 "3-2-1" proposal, included as a degenerate configuration (§3.6). Cell values follow the verbatim specifications in Highley et al. [2], the Substack series [9, 10, 3], and Highley's 3-2-1 reaction [8].

The variants trade off explainability, anti-tanking strength, and per-series tanking exposure differently. Simple COLA [10] replaces the lottery with a deterministic ordering on drought, extending eligibility to first-round losers; Simple Lottery COLA layers the pre-2019 NBA weighted odds on top of Simple's drought ordering. Classic COLA is the original variant of Highley et al. [2] and the one our manipulation bound (§4) analyzes. Countdown COLA [10] uses the McCarty number (drought $\times$ wins) and a nested-pool Monte Carlo draw that guarantees no team falls more than four spots below its priority rank. Capped COLA [3], introduced after the NBA's reported reluctance to adopt a flexible lottery line, adds a wins-based increment, a stockpile cap (MAX $= 150$), and a six-step playoff diminishment schedule; the cap bounds the per-series tanking incentive (Lemma 3). The 2026 "3-2-1" proposal [8] sits at a degenerate corner: dials Δ, ρ, C are vacuous, S collapses to single-season, and only E (a tiered 16-team pool) and W (a tiered ball allocation) carry the proposal's design (§3.6). [**Q-HIGHLEY:** Beckett COLA is used as an empirical anchor in §5 but does not appear in Table 1; supply its 7-tuple to add it to the map, or we label it an off-family reference point alongside Countdown.]

3.4 Dial-to-Property Mapping

The configuration of Definition 2 reduces variant comparison to tuple comparison. The comparison is informative only when each dial corresponds to a downstream property. Four properties dominate the adoption

Dial	Simple	Simple Lottery	Classic	Countdown	Capped	3-2-1
E	22 (non-playoff + R1 losers)	22 (same)	14 (non-playoff)	22 (same)	exclusion pool	tiered 16
Δ	drought $+1$	drought $+1$	$\alpha = 1{,}000$	McCarty $d \cdot w$	wins w (play-in & below)	none
ρ	reset on series win	reset on series win	5-step, two channels	reset on advancement	6-step + 5-step, two channels	none
W	deterministic by drought	pre-2019 odds top-14, deterministic rest	L_i/P for picks 1–4, rank for 5–14	nested-pool MC, tickets $(6, 5, 4, 3, 2)$	L_i/P for picks 1–5, rank for 6+	tiered balls $(2/3/2/1)$
C	∞	∞	∞	∞	$\text{Max} = 150$	n/a
S	reset-on-event	reset-on-event	unbounded multi-year	reset-on-event	bounded multi-year	single-season
T	most wins	subsumed by W	subsumed by W	subsumed by W	most wins (at cap)	subsumed by W

Table 1: The seven dials, evaluated for each variant. Dial labels: E eligibility, Δ increment, ρ diminishment, W lottery weighting, C cap, S carry-over scope, T tiebreak (per Definition 2). The 3-2-1 proposal sits at a degenerate corner: Δ, ρ, C are vacuous and S is single-season, leaving only E and W as active dials.

discussion: anti-tanking strength (within-season manipulation foreclosed), weak-team rebuild help (how reliably the worst teams accumulate priority), complexity (legibility for fans and media), and predictability (multi-year forecastability for GMs and owners). Table 2 summarizes the mapping.

Three dials (E, Δ, C) bound the manipulator's pool share $p_{\max}$ and the size of any pool perturbation: a larger eligible set yields a larger total pool P, the increment Δ bounds per-team per-season accumulation, and the cap C bounds the index outright. Two dials (ρ, S) govern multi-year accumulation: diminishment determines how quickly an index decays after success; carry-over scope determines whether accumulation persists at all (the dial whose non-single-season value escapes the Banchio and Munro [1] impossibility). The remaining two (W, T) translate index dynamics into draft positions. A league that finds Classic too unbounded can move C to a finite value (the Capped move). A league that finds multi-year scopes administratively heavy can move S to single-season (the 3-2-1 move). The choice among variants becomes a choice among dial settings, not among monoliths.

3.5 Incentive Compatibility

Highley et al. [2] establish incentive compatibility (IC) for Classic COLA under five assumptions:

1. **Playoff primacy:** a team's value of advancing in the playoffs exceeds any plausible draft-pick gain.
2. **Playoff advancement:** within the playoffs, advancing further is preferred to losing earlier, controlling for opponent strength.
3. **Picks 5–14 negligible at the margin:** no team profits from manipulating its position among those slots through within-season effort.
4. **Pool manipulation negligible:** no team can meaningfully shift its expected draft position by altering which other teams are in or out of the pool.
5. **No traded-pick protections:** pick trades do not interact with diminishment to create manipulation incentives.

Dial	Anti-tanking strength	Weak-team rebuild help	Complexity	Predictability
E	Sets the pool P in Theorem 2; narrower pools admit larger gain	Determines which weak teams accumulate priority	High if simple count, low if exclusion-rule	Stable if record-based; less so if exclusion-rule
Δ	Bounds the per-season marginal-decision gain	Flat α rewards uniformly; w-based rewards within-season effort	High if flat, lower if formulaic	High if record-based
ρ	Cliff size at each round transition; controls repeat-tanking returns	Restores priority to teams missing the playoffs again after success	Medium	Medium
W	Spread across rank positions; decouples position from index in part	Tilts allocation toward worst-record teams	High if deterministic, low if Monte Carlo	High if deterministic, lower if random
C	Bounds $p_{\max}$ and the per-series cost (Lemma 3)	Low cap compresses the priority signal	Medium	High once known
S	Escapes the Banchio and Munro [1] impossibility via multi-year scope	Multi-year scope rewards persistent weakness	Medium	Lower under multi-year scopes (legal/contracting)
T	Edge case for W-deterministic variants	n/a	High if simple	High

Table 2: Dial-to-property mapping. Cell entries describe the form the dial's influence takes; the dials and the variants of Table 1 together specify how to move from one league-relevant property to another by altering one or more dial values.

Assumptions (1)–(3) and (5) hold by construction or by short argument. Assumption (4) is supported in Highley et al. [2] by a 1,000-season simulation observing a maximum gain of roughly 3%. §4 converts that observation into a closed-form bound for the Classic configuration. [**Q-HIGHLEY:** This IC argument and the §4 bound are established for Classic (the top-four lottery). Does the recommended full-depth $\gamma = 1$ preserve assumptions (1)–(5) and the bound, or does lotterying every pick require re-deriving them? Our engine manipulation measurement for $\gamma = 1$ stays under $\sim 4\%$, but that is empirical, not a proof.]

3.6 The 3-2-1 Proposal as a Degenerate Configuration

The NBA's 2026 "3-2-1" proposal, slated for an owners' vote at the end of May 2026, expands the lottery to sixteen teams across four tiers with ball counts $(2, 3, 2, 1)$ per tier. In the configuration of Definition 2: Δ, ρ, C are vacuous; S is single-season; E is the tiered 16-team pool; W is the tiered ball allocation; T is subsumed by the random draw. In Highley [8]'s framing, the mechanism is "essentially the uniform lottery but with an adjustment for the play-in and a punishment for the bottom 3 teams." Because the worst three teams get fewer balls than the middle seven, the proposal does not satisfy the premise of the Munro-Banchio counterexample (worse teams have strictly better odds), so it sidesteps the impossibility by abandoning the rationale for differential odds. COLA takes the opposite route: it shifts the priority basis to multi-year playoff history through dial S.

4 Pool Manipulation Bound

4.1 Setup

Let $P = \sum_{i=1}^{n} L_i$ denote the total lottery pool, where L_i is team i's lottery index and n is the number of lottery-eligible teams. A team's probability of winning the first pick is $p_i = L_i/P$.

The *pool manipulation* channel operates as follows. Team i deliberately loses to a high-index opponent h near the playoff boundary, causing h to make the playoffs. Team h (with index L_h) exits the lottery pool and is replaced by a lower-index team ℓ (with index $L_\ell < L_h$). The pool shrinks by $\Delta = L_h - L_\ell > 0$.

Proposition 1 (Manipulation Gain)**.** *The probability gain for team i from a pool swap of size $\Delta = L_h - L_\ell$ is $G_i = L_i \cdot \Delta/[P \cdot (P - \Delta)]$.*

Proof. Before manipulation, $p_i = L_i/P$. After the swap, the new pool is $P' = P - \Delta$ and $p'_i = L_i/P'$. The gain is $G_i = p'_i - p_i = L_i\big(1/(P-\Delta) - 1/P\big) = L_i \cdot \Delta/[P(P-\Delta)]$. □

When $\Delta \ll P$ (typically $\Delta/P \approx 4\%$ in our backtest), the gain factors into the manipulator's pool share (p_i) and the fractional pool change (Δ/P). Both have structural ceilings, which we exploit below.

Indexing convention. The bound uses $L_i = T_i \cdot \alpha$, where T_i is team i's consecutive non-playoff seasons. For Classic COLA, this is exact for any team starting from a stockpile reset (championship or top-5 lottery pick) and is the steady-state attractor under the diminishment schedule. A general bound handling ticket inheritance under the full diminishment dynamics is future work.

4.2 Conditioned Upper Bound

Definition 3 (Structural Anti-Correlation)**.** *Let $T_{\max}$ denote the maximum consecutive non-playoff years for any team, and $T_{boundary}$ the maximum consecutive non-playoff years for a team near the playoff boundary (a team that could plausibly be pushed into the playoffs by a single loss). The index distribution exhibits* structural anti-correlation *when $T_{boundary} \ll T_{\max}$: teams with the highest indices are far from the playoff boundary and thus cannot be the target of pool manipulation.*

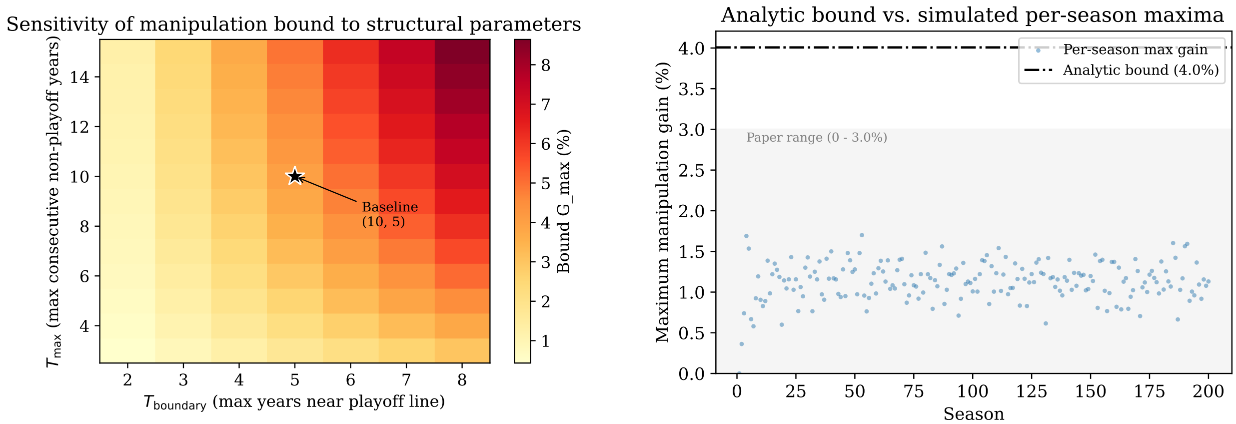
Theorem 2 (Conditioned Manipulation Bound)**.** *Under structural anti-correlation with parameters $T_{\max}$, $T_{boundary}$, and average index multiplier $\bar{k} = \bar{L}/\alpha$, where α is the annual index increment, the maximum manipulation gain satisfies*

$$G_{\max} \leq \frac{T_{\max} \cdot (T_{boundary} - 1)}{(T_{\max} + n - 1) \cdot n \cdot \bar{k}}.$$

Proof sketch. The gain is maximized when L_i is maximized and Δ/P is maximized. *Bounding* $p_{\max}$: the highest-index team has $L_{\max} = T_{\max} \cdot \alpha$. The minimum pool size occurs when one team has $T_{\max}$ years of accumulation and the remaining $n-1$ teams each have the minimum α: $P_{\min} = (T_{\max} + n - 1) \cdot \alpha$. Thus $p_{\max} \leq T_{\max}/(T_{\max} + n - 1)$. *Bounding* Δ/P: the team being pushed out has at most $T_{\text{boundary}} \cdot \alpha$ tickets, and the team replacing it has at least α, so $\Delta_{\max} = (T_{\text{boundary}} - 1) \cdot \alpha$. In steady state, $P \approx n \cdot \bar{k} \cdot \alpha$. Combining via the first-order approximation yields the stated bound. □

Empirical calibration: We instantiate with $T_{\max} = 10$ (typical-case ceiling on consecutive non-playoff years across teams in our 26-season backtest), $T_{\text{boundary}} = 5$ (conservative ceiling for a near-boundary team), $n = 14$, and $\bar{k} = 3.1$ (the original paper's observed carry-over). These yield

$$G_{\max} \leq \frac{10 \cdot 4}{23 \cdot 14 \cdot 3.1} \approx 4.0\%,$$



(a) Bound vs. $T_{\max}, T_{\text{boundary}}$.

(b) Analytic bound vs. simulation.

Figure 1: Manipulation bound. (a) Sensitivity to the structural-anti-correlation parameters; the bound is most sensitive to T_{boundary}, and stays under 5% for $T_{\text{boundary}} \leq 5$. (b) Analytic bound (red) vs. the empirical distribution of manipulation gains across 200 Bradley-Terry-simulated seasons; the bound contains all simulated outcomes with margin.

which contains the simulation-observed maximum of 3.0% across $1{,}000$ seasons [2]. An independent 200-season Bradley-Terry simulation we ran confirms the bound (max gain 1.70%). The 26-season dataset contains one outlier (the Sacramento Kings' 16-year drought, 2006–07 through 2021–22). Setting $T_{\max} = 16$ loosens the bound to roughly 5.1%, still above the empirical extremes; we report the typical-case bound ($T_{\max} = 10$) as the headline because the next-longest droughts are Minnesota at 13 years and Phoenix/Charlotte at 10.

4.3 Sensitivity, Coalitions, and the Pick-Value Gap

Figure 1a shows T_{boundary} as the dominant lever: under pessimistic assumptions ($T_{\text{boundary}} = 7$), the bound stays under 7%; under realistic conditions ($T_{\text{boundary}} \leq 5$), under 5%. T_{boundary} is also the empirical quantity most amenable to direct measurement.

Coalitional manipulation does not amplify individual gains. Joint gain for a coalition $\{A, B\}$ is $(L_A + L_B) \cdot \Delta/[P(P - \Delta)]$. At 30% joint pool share, joint gain remains $\leq 1.3\%$ and per-member gain cannot exceed the highest-index individual's. Coordination costs (detection, trust, sequential uncertainty) further dampen the channel.

The bound addresses the *probability* term of the manipulation incentive. The *value* term (the pick-value gap $d_k - d_{k+1}$) has no formal ceiling in the original specification, though it is bounded for any realistic draft class. Fully closing assumption (4) would also require a bound on $d_k - d_{k+1}$ via the distribution of draft-eligible prospect quality; we leave this to future work.

4.4 Per-Series Bound for Capped COLA

Capped COLA adds a fourth manipulation channel: tanking through the playoffs (a team within the cap declining to advance, to keep its stockpile). The cap converts the playoff-diminishment schedule from a fraction-of-current-stockpile cost (unbounded as stockpile grows) to a fraction-of-MAX cost (bounded by construction).

Lemma 3 (Per-Series Tanking Cost). *Let* MAX *be the Capped* COLA *stockpile cap. For any team entering the playoffs with $L_i \leq$* MAX*, the maximum reduction in L_i attributable to winning (rather than losing) any* *single playoff series is bounded by:*

$$\Delta L^{\max}_{series} = \begin{cases} 0.3 \cdot \textsc{Max} & \textit{play-in advancers in round 1,} \\ 0.2 \cdot \textsc{Max} & \textit{any other series advancement.} \end{cases}$$

At $\textsc{Max} = 150$*, the bound is* 45 *tickets for the play-in case and* 30 *tickets for any subsequent series.*

Proof. Reading off Capped COLA's six-step diminishment schedule, the per-series increments in the diminishment fraction $\Delta\rho$ are: play-in advancer round 1 ($\rho_{\text{lose}} = 0.1$, $\rho_{\text{advance}} = 0.4$, $\Delta\rho = 0.3$); standard round 1 ($0.2 \to 0.4$, $\Delta\rho = 0.2$); round 2 to conference finals ($0.4 \to 0.6$); conference finals to NBA finals ($0.6 \to 0.8$); finals to championship ($0.8 \to 1.0$). All non-play-in transitions yield $\Delta\rho = 0.2$. The cost in tickets is $\Delta\rho \cdot L_i^{\text{pre}} \leq \Delta\rho \cdot \textsc{Max}$, with equality at the cap. □

The bound makes the playoff-tanking incentive transparent: at $\textsc{Max} = 150$, no single series win ever costs more than 45 tickets, roughly a year and a half of accumulation for a team near the wins-increment ceiling. The cumulative regret for a finals-losing team at the cap is $0.8 \cdot \textsc{Max} = 120$ tickets, decomposed per Lemma 3.

Corollary 4 (Cap-Cancels-Out for Capped Configurations)**.** *For a Capped* COLA *configuration with eligibility pool* E *of size* $|E|$*, cap value* C*, and per-series-cost upper bound* $\kappa \cdot C$ *(where* $\kappa = 0.3$ *in the play-in worst case of Lemma 3 and* $\kappa = 0.2$ *for any subsequent series), the worst-case manipulation-gain bound for a single-series pool perturbation is*

$$\Delta p_{capped} \;\leq\; \frac{\kappa \cdot C}{C \cdot |E|} \;=\; \frac{\kappa}{|E|}.$$

The cap value C *drops out. The manipulation-gain bound depends only on* κ *and* $|E|$*.*

Proof. The maximum index a single team can hold under Capped COLA is C, and the worst-case pool of $|E|$ such teams is $C \cdot |E|$. The maximum single-series perturbation in a team's index from the pool is $\kappa \cdot C$ tickets, by Lemma 3. Substituting into the gain expression $\Delta = (\text{perturbation})/(\text{pool})$ yields $\kappa \cdot C/(C \cdot |E|) = \kappa/|E|$. □

The cap controls per-team accumulation speed and per-series tanking cost (Lemma 3); it does not control pool-wide manipulation resistance, which is set by $|E|$. A policymaker can tune C on accumulation-speed and per-series-cost preferences without affecting the manipulation-gain ceiling. At Capped's published defaults ($|E| = 22$, $\kappa = 0.3$ play-in or 0.2 standard), the bound is $\Delta p_{\text{capped}} \leq 0.3/22 \approx 1.36\%$ play-in and $0.2/22 \approx 0.91\%$ for any subsequent series.

4.5 Three Closures of Assumption (4)

The pool-manipulation gap admits three closure strategies. *Empirical:* Highley et al. [2] simulate $1{,}000$ Bradley-Terry seasons and observe maximum gain $\sim 3\%$ (statistical, no parameter-explicit ceiling). *Behavioral:* the companion formal-proof video to [2] introduces $G >$ expected pool-manipulation gain (where G is the value of a regular-season win), eliminating the channel via preference structure. *Analytic (this paper):* Theorem 2 gives a closed-form bound conditioned on structural properties of the index distribution, parameter-explicit and admitting sensitivity analysis (Figure 1a). The three compose; any practical deployment can rely on whichever best matches the available evidence.

5 Empirical Validation

We instantiate the configuration of §3 in the full ZenGM Basketball-GM engine. §5.1 discloses the simulation model and why it is the right reference for a draft-reform question; §5.2 describes the weighting sweep, the gradient metrics, and the manipulation probe; §5.3 reports the headline ($\gamma = 1$ full-depth proportional weighting closing the single-season tanking gradient, help directed by drought, equity channel-limited at the outcome, and simulated manipulation confirming Theorem 2) together with the run's limits; §5.4 carries the 26-season historical backtest.

5.1 Simulation Model

The ZenGM Basketball-GM engine implements all five COLA variants and the standard NBA lottery; we use it as our reference implementation and run it headless with the full per-game simulation active. The engine generates rosters and contracts, simulates each game, lets AI general managers draft and trade, and applies city-size effects; team strength emerges from the rosters the engine actually plays, not from a hand-specified law of motion. This is the realism the draft-reform question requires: tanking, rebuild help, and pool manipulation are all consequences of how a drafted player changes the team that drafts him, so the channel from draft pick to standings has to be the engine's own, not a stipulated one. An earlier pass synthesized regular-season outcomes (strength-weighted win records and a single-elimination bracket under a strength Markov model) because ZenGM's browser-coupled `createLeague()` path does not port to Node; the present study drives the engine through its headless path instead. The ZenGM lottery code is unchanged: draws use `draft.genOrder` and per-team `cola` updates from `cola.ts`.

We report uncertainty at the replicate level (effective sample size 48 per mechanism; see the Simulation-parameters note), so within-league season correlation does not inflate significance. Each claim carries a 95% confidence interval: a near-zero claim rests on an interval that includes zero, a discriminating claim on one that excludes it.

5.2 Testbed Setup

A channel probe (§5.3) first establishes how much a draft pick is worth in the engine, so the rest of the study can be read against the right magnitude. The main experiment then varies one dial of Definition 2, the lottery weighting W, as a single-knob family, holding the increment, diminishment, cap, carry-over scope, and tiebreak at their Classic settings and standardizing eligibility at the 22-team pool across all mechanisms, so the weighting contrast is clean. (Standardizing E at 22 makes the empirical "Classic" differ from the canonical Classic of Table 1, which sets $E = 14$; a confirmatory rerun at $E = 14$ is pending.) The full draft order over the eligible pool (the 22 teams that did not win a playoff round) is drawn with probability proportional to the multi-year index raised to a power, L_i^γ: $\gamma = 0$ is a flat lottery (equal chance among the eligible), $\gamma = 1$ is proportional to drought (full-depth proportional weighting), and $\gamma > 1$ concentrates odds on the most-droughted. Drawing every pick this way removes the record-order tail that Classic leaves on picks 5 and beyond, which is the residual single-season tanking channel. The coarse pass sweeps $\gamma \in \{0, 1, 2, 3\}$ with Classic, Countdown, and Beckett as named reference points, each over 48 independent leagues of 15 consecutive seasons. [**Q-HIGHLEY:** The two record-based mechanisms, the NBA's 3-2-1 and the traditional weighted lottery, are not yet in the sweep; we will add them to complete the own-metric comparison, and expect each to show its help gap and tanking gradient nearly equal (3-2-1's negative, since it gives the worst teams fewer balls).]

We score three legs. *Anti-tanking (leg a)* is the single-season tanking gradient: the gap in expected pick between the worst-record and fifth-worst-record team when teams are ranked by the current season's record. A positive value is the tanking reward (losing one more game this season buys a better pick); the target is about zero. *Equity (leg b)* is read two ways at the outcome level: the drought tail (the 90th percentile of years since a team's last conference-final appearance, lower is better) and the recovery rate (among team-seasons in a drought of at least three years without a conference final, the share that reach the conference finals within the next three seasons, higher is better), both compared against the $\gamma = 0$ flat baseline. A complementary drought-to-pick help gradient measures the affirmative side of the design at the pick level: the slope of assigned pick on accumulated drought-years among pool teams, where a negative slope means a longer-drought team drafts earlier. *Manipulation (leg c)* is measured, not assumed: we simulate the pool-swap loophole on the engine's realized pre-draft index (a team near the playoff boundary is pushed into the playoffs, shrinking the pool) and record the realized gain in pick-1 probability, in both the paper's 14-team non-playoff framing and the engine's actual 22-team pool, to compare against Theorem 2's $\sim 4\%$ ceiling. Per-leg quantities are aggregated as the mean across the 48 leagues with the standard error of that mean, so within-league season correlation does not inflate significance.

5.3 Results

The draft channel is modest. The channel probe assigns picks at random and measures the effect on later standings. A first-overall pick is worth about two to three wins over a thirtieth pick at its peak (channel probe, $n = 12$); the all-teams estimate sits about 5.2 standard deviations from zero and the weak-team estimate about 4.6, and the assigned pick stays orthogonal to current record (contemporaneous slope near zero), so the effect is causal but small. This sets the scale for the rest of the study: which team a mechanism helps moves at the pick level, but how far that help carries to the standings is limited by how little one pick is worth. The random arm is also the uniform-lottery baseline the COLA weightings must beat on equity.

Single-season tanking is the discriminating axis (leg a). Table 3 reports the tanking gradient by weighting. Classic leaves a $+2.48 \pm 0.11$ pick reward, about twenty-two standard deviations from zero: because Classic lotteries only the top four picks and orders the rest by record, a structure Highley et al. [2] keeps deliberately to line up with the current NBA lottery; the by-record tail on picks 5 and beyond is where a worse current record buys a better expected pick. Drawing every pick by the multi-year index closes that tail, at the cost of that top-four structure. Flat ($\gamma = 0$) and full-depth proportional ($\gamma = 1$) weightings both land within noise of zero (-0.64 ± 0.35 and $+0.29 \pm 0.33$); steeper shapes re-couple picks to current record (a mild reward returns by $\gamma = 2$, $+0.78 \pm 0.31$), and the named anchors over-correct into a penalty where losing actively costs draft position (Countdown -2.85 ± 0.26, Beckett -0.76 ± 0.24).

Help is directed by drought (leg b, pick level). The complementary gradient asks the affirmative question: does a longer-drought team draft earlier than a briefly-bad one? Table 4 reports the slope of assigned pick on accumulated drought-years among pool teams. Full-depth proportional weighting separates here: each extra drought-year moves a $\gamma = 1$ team about one pick earlier (-0.94 ± 0.02), so a six-year-drought team drafts around ninth against around fourteenth for a one-year team. The flat lottery directs no help by drought (-0.01 ± 0.02: every pool team is equal); Classic directs help but pays the single-season tanking reward of leg (a) alongside it (-0.62 ± 0.02). Only $\gamma = 1$ both aims help sharply at the long-droughted and holds the single-season tanking gradient at about zero. [**Q-HIGHLEY:** Adopt the own-metric scoring you proposed in place of (or alongside) this OLS slope? Rank each mechanism by its own priority metric (record for the NBA lottery and 3-2-1, the index for COLA) and report the expected-pick gap between its highest-priority and fifth-priority team; under that scoring the help gaps are Classic $+1.10$ and $\gamma = 1$ $+1.55$, and for any record-based mechanism the help gap equals the tanking gradient, which is the cleanest statement of why only COLA separates the two.]

weighting	tanking gradient (a)	drought p90, yrs (b)	recovery rate (b)
Classic (status quo)	$+2.48 \pm 0.11$	11.0	0.202
$\gamma = 0$ flat	-0.64 ± 0.35	10.8	0.206
$\gamma = 1$ proportional	$+0.29 \pm 0.33$	10.9	0.200
$\gamma = 2$ steep	$+0.78 \pm 0.31$	11.0	0.205
$\gamma = 3$ steeper	$+0.98 \pm 0.32$	11.0	0.200
Countdown	-2.85 ± 0.26	10.8	0.214
Beckett	-0.76 ± 0.24	10.9	0.204

Table 3: Lottery-weighting sweep, 48 leagues $\times$ 15 seasons per mechanism (means $\pm$ standard error across the 48 leagues). Leg (a) is the single-season tanking gradient (worst-versus-fifth-worst expected-pick gap ranked by current record; about zero is the target, a positive value is a tanking reward). Leg (b) is equity at the outcome: drought tail (90th percentile of years since a conference final, lower is better) and recovery rate (share of teams in a $\geq$ 3-year drought that reach the conference finals within three seasons, higher is better). The flat ($\gamma = 0$) arm is the uniform-lottery equity baseline.

mechanism	picks per drought-yr	1–2 yr	3–5 yr	6+ yr
$\gamma = 1$ proportional	-0.94 ± 0.02	14.2	11.7	8.8
Classic	-0.62 ± 0.02	13.5	11.6	9.6
$\gamma = 0$ flat	-0.01 ± 0.02	11.3	11.5	11.6

Table 4: Drought-to-pick help gradient among pool teams (means $\pm$ standard error across the 48 leagues; drought is the pre-draw multi-year index, about one unit per year out of contention). A more negative slope directs help more sharply to the long-droughted; the mean assigned pick should fall from the 1–2 year bucket to the 6+ bucket if the mechanism rewards sustained drought. Only $\gamma = 1$ produces a steep negative slope while holding leg (a) near zero.

Equity is channel-limited at the outcome. The help that $\gamma = 1$ directs at the pick level does not translate into a proportional outcome gap. Across mechanisms the drought tail (about 11 years) and the recovery rate (about 0.20) barely move (Table 3, last two columns): the equity nulls rule out large differences between weightings, not small ones. The probe explains why. Because one pick is worth only about two to three wins, a large difference in which droughted team receives the top pick still produces only a small difference in which team returns to contention. The weighting is therefore a genuine help dial at the pick level; it is the draft channel itself, not the weighting, that limits how much that help shows up in return-to-contention rates. This is the strongest counterargument to a strong equity claim, and we state it as a limit rather than work around it: at this horizon and sample, we can show $\gamma = 1$ aims help correctly, not that it returns droughted franchises to contention at a measurably higher rate.

Manipulation, measured, confirms Theorem 2 (leg c). Rather than assume the pool-manipulation channel is negligible (assumption (4) of §3.5), we simulate it on the engine's realized pre-draft index. Table 5 reports the worst-case realized gain in pick-1 probability from a single pool swap. The recommended $\gamma = 1$ stays under Theorem 2's $\sim 4\%$ ceiling: its worst-case realized gain is at most 2.67% in the paper's 14-team non-playoff framing and at most 1.54% in the engine's actual 22-team pool, with a 95th percentile near 1.2%. The flat lottery $\gamma = 0$ is the lone exception, with a single worst season at 4.36% (its uniform odds give every eligible team a stake in the pool size), one more reason to prefer $\gamma = 1$ to flat. The theorem's structural-anti-correlation premise (Definition 3) holds in the engine: drought and current record are anti-correlated, so the best-record non-playoff team, which is the manipulation target, carries only about 60% of the pool's peak index (boundary-to-peak ratio about 0.62 under $\gamma = 1$), and the swappable pool change

mechanism	max gain, 14-pool	max gain, 22-pool	boundary index / pool peak
Classic	2.82%	2.20%	0.57
$\gamma = 1$ proportional	2.67%	1.54%	0.62
$\gamma = 0$ flat	4.36%	1.89%	0.58

Table 5: Realized pool-manipulation gain (leg c) versus Theorem 2's $\sim 4\%$ bound. Gain is the worst-case realized pool-swap advantage in pick-1 probability, simulated on the engine's pre-draft index, in the paper's 14-team non-playoff pool and the engine's 22-team COLA pool. The boundary-to-peak ratio (well below 1) is the empirical form of the structural-anti-correlation premise (Definition 3): the manipulation target, a near-boundary team, sits far below the highest-index team. $\gamma = 1$ clears the bound on both pool framings; the flat lottery's single worst season (4.36%) is the lone excursion above it.

is small. The engine's 22-team pool, which a team can leave only by winning a playoff round, is about half as manipulable as the 14-team framing. This is the empirical confirmation of the analytic bound: the closed-form $\sim 4\%$ is an envelope, and the realized opportunity inside the engine sits well below it.

What the run does and does not establish. The discriminating result is firm: Classic's $+2.48$ tanking gradient is about a twenty-two-standard-deviation effect. The headline fix, $\gamma = 1$ closing that gradient to about zero, holds a 95% confidence interval of $[-0.37, 0.95]$ that includes zero (statistically no tanking reward), while $\gamma = 2$'s and $\gamma = 3$'s intervals exclude zero (a measurable reward returns), and the Classic-minus-$\gamma = 1$ contrast is Welch $t = 6.35$, $p < 10^{-6}$, Cohen's $d = 1.30$. Four limitations remain. The realized-recovery null (equity flat across mechanisms) is a substantive finding from the modest draft channel, not a power problem: one pick is worth only about two to three wins, so even a large pick-level help difference moves return-to-contention little. Manipulation is the realized opportunity envelope from the engine's index distribution, not a behavioral agent that actually throws games, so it bounds the temptation rather than the take-up; its worst case is a noisy single-season tail, and the 95th percentile (about 1.2% for $\gamma = 1$) is the steadier read. The 15-season horizon partly censors the drought tail (the 90th percentile near 11 years sits close to the ceiling the horizon allows). The results use one engine, unpaired across mechanisms; a common-random-numbers design (shared seeds across weightings) would tighten the contrasts further.

The anti-tanking weighting and its tradeoff ($\gamma = 1$). Across the three legs, one weighting clears all of them. On anti-tanking (leg a), $\gamma = 1$ holds the single-season gradient at $+0.29 \pm 0.33$, indistinguishable from zero, against Classic's $+2.48$. On help (leg b), $\gamma = 1$ directs draft priority to the long-droughted most sharply of any mechanism at the pick level, about one pick earlier per drought-year, with realized recovery channel-limited and tied to the flat baseline. On manipulation (leg c), its realized gain is at most 2.7% in the paper's pool and at most 1.5% in the engine's, under the $\sim 4\%$ bound. The flat lottery ($\gamma = 0$) is anti-tanking but aims no help; Classic aims help but pays the single-season tanking reward. The change $\gamma = 1$ asks of Classic is one dial on the weighting: keep the lottery odds proportional to drought, and draw every pick that way. That move removes the record-order tail that makes tanking pay, at no measured cost to rebuilding or to manipulation resistance. It is not free, however: drawing every pick by the index gives up the top-four lottery that Classic keeps to line up with the current NBA system, a deliberate familiarity choice in Highley et al. [2]. We therefore report $\gamma = 1$ as the weighting that clears the anti-tanking test, presented as a tradeoff against that familiarity rather than a strict improvement. [**Q-HIGHLEY:** Two questions decide how we present this. (1) Is the top-four lottery limit purely practicality and familiarity, or are there technical reasons (incentive compatibility, the carry-over and diminishment dynamics, or Theorem 2's manipulation bound, all derived for the top-four Classic) that full-depth weighting would break? (2) If the limit is mainly familiarity, do you prefer the middle option, keeping the four-pick lottery and ordering picks 5–14 by the index, which closes the same tail while preserving the NBA-familiar structure? We have not yet run that middle option.] **[Q-HIGHLEY:** Naming: carry $\gamma = 1$ as a setting of Classic COLA, or name it on its own? "Classic" is this paper's taxonomy label; Highley et al. [2] calls the base mechanism COLA.]

5.4 Historical Backtest

The backtest computes COLA state for every team across 26 NBA seasons (1999–2025) using verified Basketball Reference data: 775 team-season records, 776 first-round draft picks, and 55 lottery-day-attributed pre-draft pick trades. The engine separates pre-draft state (Phase A, used for draft-probability calculations) from post-draft diminishment (Phase B, carried forward to the next season). Cold-start effects stabilize by approximately season 5; we report aggregates over 21 stable seasons (2005–2025). The backtest is *static*: it applies COLA rules to actual historical outcomes. Under a real COLA regime, draft outcomes would differ in ways that change future drought lengths and indices. We caveat the comparative cross-variant claims accordingly. An audit of all 364 lottery picks (14 picks $\times$ 26 seasons) identified 8 instances where the draft-day holder differed from the lottery-day holder via post-lottery trades; COLA uses *lottery-day* attribution.

The 2013–14 through 2016–17 Philadelphia 76ers are the canonical strategic-tanking sequence in modern NBA history: four consecutive top-3 lottery outcomes (picks #3, #3, #1, #3) acquired through deliberate multi-season non-competitiveness. Under the current weighted lottery, each tanked season yielded a top-tier pick. Under Classic COLA, graduated diminishment makes the same sequence self-defeating. Philadelphia's Classic COLA trajectory: 2,750 in 2013–14 (rank 8/14) $\rightarrow$ 2,375 in 2014–15 (9/14) $\rightarrow$ 2,188 in 2015–16 (10/14) $\rightarrow$ 1,000 in 2016–17 (14/14). Each top-3 pick triggers graduated diminishment ($\#1$: full reset; $\#3$: 50% reduction), leaving Philadelphia at the bottom of the lottery pool by the fourth tanking season despite a 10–72 record. The Sacramento Kings hold the top Classic COLA index throughout this period, climbing from 7,250 in 2013–14 to 10,250 in 2016–17 via organic non-competitiveness. The Classic diminishment schedule ($\{1.0, 0.75, 0.5, 0.25\}$ for picks #1–4) dominates the $\alpha = 1{,}000$ annual accumulation; a team that receives a top-3 pick in year t cannot re-enter the top of the lottery in year $t + 1$ through tanking alone.

The Capped cap calibration sweep over MAX $\in \{75, 100, 125, 150, 175, 200\}$ across the 21 stable seasons replicates the published default of MAX $= 150$ [3]: at MAX $= 150$, a mean of 1.62 teams sit at the cap (range 0–3), matching the design target of fewer than two teams at the maximum on average. Sub-100 caps over-compress the rank-1-vs-rank-5 separation; caps above 175 leave fewer than one team at the cap on average. The static backtest is a second-tier validation alongside the headline full-engine result: it confirms the qualitative comparative-statics the full-engine study reports under closed-loop dynamics. Trade handling is a practical implementation question for any deployment; following Highley [11], we recommend original-holder diminishment paired with a strengthened Stepien Rule.

6 Discussion and Open Directions

Theorem 2 bounds Classic COLA's open assumption (4) at approximately 4% under empirically calibrated parameters, with explicit sensitivity to $T_{\max}$, T_{boundary}, n, $\bar{k}$. Lemma 3 bounds Capped's per-series exposure. Combined, the four manipulation channels under Capped (single-season tanking, pool manipulation, multi-year strategic tanking, per-series playoff tanking) compose to a small joint exposure relative to the status-quo lottery, where repeat tanking has memoryless returns.

6.1 3-2-1 as 2026 Stopgap, COLA as 2029 Candidate

We read 3-2-1 as a short-horizon stopgap: it accepts the Munro-Banchio tradeoff by flattening odds, with the bottom three teams given fewer balls than the middle of the pool. The proposal's three-year re-evaluation provision reopens the design discussion in 2029. COLA is the natural 2029 candidate on a different axis: dial

S (carry-over scope) shifts the priority basis to multi-year playoff history, an option neither 3-2-1 nor any single-season mechanism considers. 3-2-1 sacrifices differential reward for non-tanking; COLA preserves differential reward by relocating it to a non-manipulable basis. The 2029 discussion benefits from having both options analytically characterized.

6.2 Dial-Space Adoption Tradeoffs

Each dial matters more to some audiences than others. *Fans and media* care most about E and W: the current 14-team weighted lottery fits a 30-second broadcast graphic, while Capped's exclusion-rule pool and Countdown's nested Monte Carlo do not. *General managers and owners* care most about Δ and ρ: a multi-year rebuild needs a forecastable trajectory. Classic's flat $\Delta = \alpha$ and graduated ρ are the most tractable; Capped's wins-based Δ and six-step ρ add complexity but remain forecastable; Countdown's $\Delta = d \cdot w$ multiplies factors that move in opposite directions for a rebuilding team and produces noisier multi-year forecasts. *League counsel* cares most about S: multi-year scopes raise questions about mid-season team relocation, ownership change, and league realignment that single-season scopes avoid. The NBA's apparent preference for single-season scopes in 3-2-1 is consistent with S being the dial under the most external pressure.

6.3 Reset-on-Championship as a Policy Lever

The configuration of Definition 2 exposes a third option for S beyond time-bounded windows and unbounded accumulation: an outcome-triggered reset (the *reset-on-event* value of the carry-over-scope dial). For league counsel, reset-on-championship bounds the carry-over horizon without imposing a calendar deadline, and the trigger event is already part of the league's annual cadence. For GMs, it preserves the multi-year rebuild trajectory until the rebuild succeeds; winning the championship resets the priority index. Event-based scopes extend the design space beyond time-based alternatives; a full-engine evaluation of the carry-over-scope dial, which the present study holds fixed while it varies the weighting, is the natural next sweep.

Open directions: three follow-on questions extend this work. (i) *Tighter manipulation bound:* the structural anti-correlation in Definition 3 is a calibrated assumption rather than a derived consequence of the mechanism. A stationarity argument bounding T_{boundary} as a function of league parameters (team count, season length, playoff structure) would close assumption (4) entirely. (ii) *Pick-value gap:* combining the probability bound with a bound on the pick-value gap $d_k - d_{k+1}$, derived from the distribution of draft-eligible prospect quality, would close the manipulation incentive in absolute terms. (iii) *Axiomatic embedding:* the lift from Coreno and Balbuzanov [4]'s axioms (RP, EF1, SP) to multi-period priority mechanisms remains open; a formal lift would embed COLA in the assignment-mechanism literature and clarify which one-period impossibilities transfer.

A reproducibility artifact accompanies this paper: an interactive backtester implementing all five variants plus the 3-2-1 proposal across 26 NBA seasons, with locked-seed Monte Carlo for Countdown probabilities [12].

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